



Catchment Wisdom Framework

From data to collective wisdom

Cross-Catchment Collective
Working framework
V1

Contents

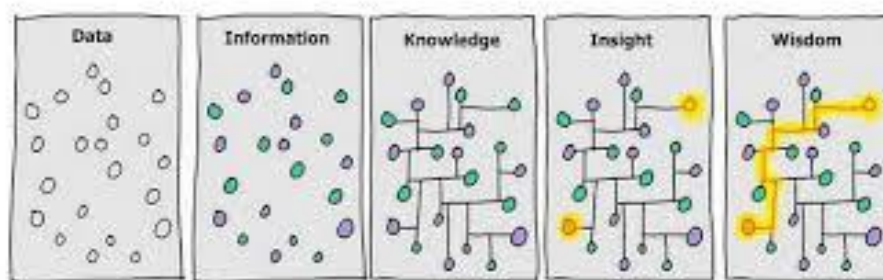
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1. Introduction and purpose

The purpose of this framework is to enable adaptive, evidence-based and community-informed decision-making across catchments, with a focus on collective learning and reflective evaluation. Monitoring and reporting serve as supporting functions, feeding into cycles of reflection, adaptation, and strategic action.

Catchment Wisdom is the collective, experience-informed guidance that partners agree upon to shape future decisions, policies, and practices across catchments.

It is when the “aha!” moments from data, science and lived experience are crystallised into **shared norms, agreements or practices** that shape how land and water are managed at a catchment scale.



[Cartoon by David Somerville, based on a two-pane version by Hugh McLeod]

The journey from data to wisdom (cartoon by David Somerville, based on a two-pane version by Hugh McLeod).

Catchment Wisdom goes beyond technical knowledge by integrating scientific evidence, local knowledge, lived experience, and institutional memory. It emerges when:

- **Data** – raw observations, measurements, records – are systematically gathered.
- **Information** is generated by processing and contextualising data for relevance.
- **Knowledge** develops when information is interpreted, compared, and validated by different actors against specific contexts.
- **Insight** arises through reflection, dialogue, and application, revealing patterns, drivers, and trade-offs.

Through co-creation and validation at catchment scale, this iterative process produces guidance that is:

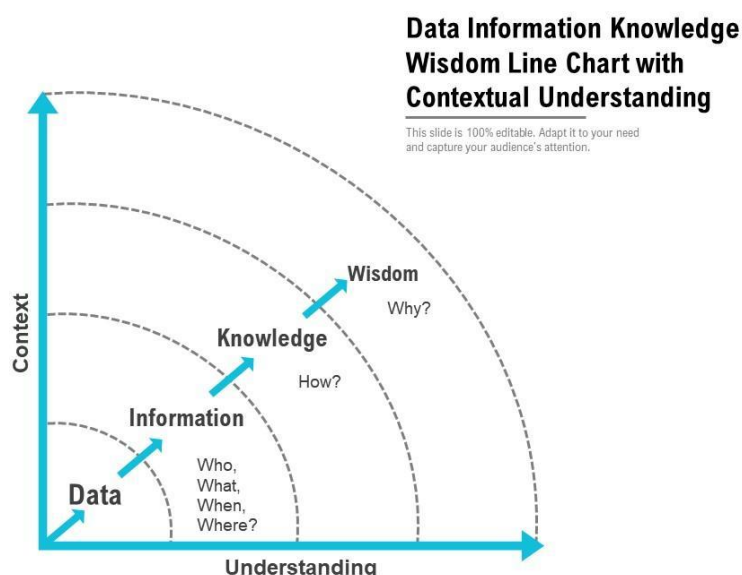
- **Collective** – shaped by multiple stakeholders (communities, government, researchers, private sector).
- **Contextual** – grounded in the unique social, ecological, and hydrological realities of the catchment.
- **Action-oriented** – translated into practical pathways for improved governance, sustainable use, and resilience.
- **Evolving** – continuously refined as new data, experiences, and innovations emerge.

In practice, Catchment Wisdom represents a shared compass: it helps partners align strategies, anticipate risks, and act collectively, ensuring that water, land, and livelihoods are managed in ways that balance equity, resilience, and sustainability.

2. Guiding principles

- **Learning first:** the system is designed to generate insight, not just track compliance.

- **Participatory:** communities, partners, and stakeholders shape what is learned, how it is interpreted, and how findings are applied.
- **Adaptive:** findings inform ongoing adaptation, not just end-of-project reporting.
- **Systems-oriented:** focus on interactions, patterns, and change across catchments rather than isolated outputs.
- **Equity and inclusion:** diverse voices, especially marginalised groups, are central to sense-making.



The DIKW progression: as context and understanding deepen, data becomes information, knowledge, and ultimately wisdom.

3. The wisdom journey

This framework makes the DIKIW journey (data → information → knowledge → insight → wisdom) visible by:

1. Asking partners to document how they transformed raw data into higher-order understanding (wisdom).
2. Using collective workshops to stitch together individual learning into system-wide wisdom.
3. Producing accessible outputs that show the value of collective evidence, not just isolated monitoring.

The flow from raw data to collective guidance can be summarised as follows:

Step	Process	Example output
1. Data	Field measurements, observations, stories	Rainfall logs, community minutes
2. Information	Data cleaned, visualised, contextualised	Graphs, maps, dashboards
3. Knowledge	Shared interpretation and validation	"Solar pumps improved access in X area"
4. Insight	Collective reflection revealing drivers	"Local leadership reduces system failures"
5. Wisdom	Agreed guidance shaping action	"Integrate traditional leaders in co-management"

Components of the wisdom journey

1. Data – raw facts from the field

- Collected by communities, partners, researchers, and technical tools.
- Examples: rainfall readings, water quality tests, farmer practices, community meeting notes.

Inputs are fragmented and raw.

2. Information – organised data

- Data cleaned, sorted, visualised, and compared.
- Examples: maps of water access points, graphs of rainfall trends, participation records.

Now accessible and usable for discussion.

3. Knowledge – shared understanding and interpretation

- Collective sense-making by partners.
- Evidence linked to lived experience and local knowledge.
- Example: “Water access improved in villages where solar pumps were installed.”

A common story begins to form.

4. Insight – new perspectives and implications

- Learning across stakeholders, revealing patterns and drivers.
- Examples: “Strong local leadership reduces breakdowns”; “Catchment-wide issues need inter-village coordination.”

Understanding deepens; choices and trade-offs become clear.

5. Wisdom – catchment-level guidance for action

- Insights translated into strategies, agreements, or policies.
- Collective wisdom applied across villages and catchments.
- Examples: “Co-management agreements must include traditional leaders”; “Scaling water-smart agriculture requires local enterprise support.”

Shared wisdom guides decisions, resource allocation, and advocacy.

4. Examples from the field

South Africa – Olifants and Letaba catchments

- **Water allocation trade-offs:** partners recognised that irrigation expansion upstream reduced downstream flow to villages. Through dialogue and shared monitoring, stakeholders agreed on seasonal restrictions and staggered pumping times to balance food production and basic needs.
- **Alien invasive plant control:** data showed water loss from invasives such as lantana. Landowners, farmers, and NGOs co-created catchment-wide clearing programmes, integrating local job creation (EPWP teams) with improved streamflow management.

Ethiopia – Upper Awash catchment

- **Flood and drought management:** communities, hydrologists, and government analysed rainfall and flow data. They co-designed seasonal flood warnings, small-scale check dams, and flexible grazing patterns.
- **Wisdom element:** insights on when and where floods strike most severely were embedded in community bylaws and district planning.

Generalised examples of catchment wisdom

- Agreed water-sharing rules between villages, farmers, and industries during drought periods.
- Catchment-wide restoration norms (e.g. no farming on steep slopes, riparian buffer zones maintained).
- Community-driven water quality monitoring that informs municipal action on pollution hotspots.
- Institutional memory embedded in bylaws, ensuring that lessons from past droughts and floods guide future planning.

5. Pathways and processes

The MERL cycle

1. Set learning priorities (annual learning agenda).
2. Collect key point-monitoring data (continuous).
3. Conduct reflective evaluations (periodic).
4. Host learning spaces (quarterly labs, annual summits).
5. Adapt strategies (feed learning into planning).
6. Share information (accessible reporting and dissemination).

Roles and responsibilities

- **Partners:** co-create indicators, collect and interpret local data.
- **Cross-Catchment Collective secretariat:** facilitate evaluation, convene learning spaces, and synthesise findings.
- **Partners and researchers:** support evidence generation, analysis, and knowledge translation.
- **Others (e.g. WRC):** engage in learning spaces and use findings to inform policy and resourcing.

Supporting processes along the way

- **Learning Labs:** partners reflect together at each step.
- **Cross-Catchment Exchanges:** insights tested and validated in different contexts.
- **Wisdom Summits:** annual synthesis into practical guidance.

6. Tools and templates

DIKIW Learning Template – Partner Learning Record

Partner name: _____ Catchment: _____

Date / period: _____ Theme / focus area: _____

1. Data – what we collected

Facts, observations, numbers, stories.

- What did we collect? (e.g. water samples, rainfall logs, meeting records, community feedback)
- Who collected it?
- When?

2. Information – how we organised it

Processed or structured data that makes sense.

- How was the data analysed, visualised, or categorised? (e.g. graphs, maps, summaries)
- What patterns or trends became visible?

3. Knowledge – what we learned together

Shared understanding from interpreting information.

- What did we conclude from the information?
- What did the community or partners say during sense-making?
- Which assumptions were challenged or confirmed?

4. Insight – what this means for us

Deeper understanding and implications.

- What new perspective did we gain?
- What is changing because of this?
- What opportunities or risks became clearer?

5. Wisdom – how we will act

Guidance for decisions, action, or policy.

- What decision or action did we take (or recommend) based on this learning?
- How does this influence our collective strategy?
- How can this be shared with others across the Collective?

6. Evidence / artefacts attached

(tick or attach)

- ☐ Photos / videos
- ☐ Maps / graphs / tables
- ☐ Meeting notes
- ☐ Stories / quotes
- ☐ Reports

7. Collective learning contribution

- How does this add to shared wisdom across catchments?
- What do we want other partners to learn from this?

Annex A – Proposed full framework structure

The following outline captures the fuller structure envisaged for this framework. Sections not yet drafted are retained here as a development roadmap.

1. Introduction

- 1.1 Purpose – from MERL to Wisdom
- 1.2 Why Catchment Wisdom matters
- 1.3 How this framework complements the MERL Guidebook and Knowledge Framework
- 1.4 Defining Catchment Wisdom
- 1.5 Guiding principles (learning first; participatory and inclusive; adaptive; systems-oriented; equity-centred)

2. Conceptual foundations

- 2.1 The DIKIW framework (data → information → knowledge → insight → wisdom)
- 2.2 Linking evidence-based planning to adaptive management
- 2.3 Wisdom as collective, experience-informed guidance
- 2.4 The relationship between MERL and Catchment Wisdom

3. The wisdom journey

- 3.1 Example template: Cross-Catchment Collective Learning Record
- 3.2 Examples from the field: Olifants, Letaba, Upper Awash
- 3.3 Generalised examples of Catchment Wisdom outcomes

4. Pathways and processes

- 4.1 Learning loops and feedback mechanisms
- 4.2 Annual learning agenda and reflection cycle
- 4.3 Learning labs and cross-catchment exchanges
- 4.4 Wisdom Summits – synthesising shared insights
- 4.5 Integrating wisdom into decision-making and planning
- 4.6 Avoiding “navel-gazing” – ensuring action and policy linkages

5. From insight to action

- 5.1 Translating wisdom into governance norms, bylaws, and plans
- 5.2 Embedding lessons in institutional systems (CMA, municipalities, traditional authorities)
- 5.3 How wisdom informs resource allocation and prioritisation
- 5.4 Dissemination and uptake strategies – CCC, WRC, etc.

6. Common data points and shared value

- 6.1 Why cross-catchment data matters
- 6.2 Common indicators and trends
- 6.3 What collective analysis offers beyond individual projects
- 6.4 Building national “wisdom datasets” – from hydrology to livelihoods

7. Linking to research and policy

- 7.1 Aligning with research frameworks (WRC, universities)

- 7.2 Integrating communal and traditional governance knowledge
- 7.3 Translating wisdom into policy recommendations
- 7.4 Feeding lessons into broader systems (Water Funds, SDG monitoring, etc.)

8. Tools and templates

- 8.1 DIKIW Learning Template (Partner Learning Record)
- 8.2 Reflection session and Wisdom Lab facilitation guide
- 8.3 Cross-Catchment Learning Summary template
- 8.4 Example synthesis brief format
- 8.5 Visual storytelling tools (maps, timelines, infographics)

9. Case examples of Catchment Wisdom

- 9.1 South Africa: Olifants-Letaba collaboration on water sharing
- 9.2 [to be developed]
- 9.3 Shared governance in communal vs. titled land systems
- 9.4 Lessons in institutionalising learning

10. Sustaining the wisdom cycle

- 10.1 Roles and responsibilities
- 10.2 Building the culture and practice of reflection
- 10.3 Ensuring continuity through leadership transitions
- 10.4 Measuring the “wisdom effect” – tracking policy or practice shifts over time

11. Annexes

- A. Glossary of Wisdom-related terms
- B. Reading list and learning network links
- C. DIKIW framework reference visuals
- D. Cross-Catchment Collective resources

Annex B – Working notes (to be co-developed)

These are unedited working notes captured during framework development, retained for reference. They are not part of the finalised framework text.

MERL is a platform for how data and information are distilled into wisdom. Key questions to work through:

- What does each catchment regard as its own “wisdom”?
- Get an example for the Rivers of the Lowveld first, then gather from everyone else.
- What could wisdom reasonably affect – e.g. taking it to the Water Research Commission?
- How do we mobilise through the CCC and share with everyone else?

On processes (otherwise this becomes navel-gazing):

- What are the common data points we are all collecting? What does the data offer us that we would not get working on our own – data analysis, trends, a systems view, and trends across multiple catchments.
- Ecosystem protections; hydrological studies; land-use impacts.

On reflecting on practice as part of wisdom:

- What have we learned about reserve determination – that it takes too long, is too complicated and too expensive, and that lawyers won’t take it on because it is too complicated? This would not come into MERL – it is part of wisdom.
- When dealing with communal rather than commercial land, traditional governance systems are very different from those with title deeds, and therefore need to be addressed differently.
- Link with the research framework and MERL – this is where the wisdom emerges; review how it refers to MERL.